

SimulIDE For Simulating Electronic Circuits

PRIYA RAVINDRAN

Hobbies are fascinating stuff. Be it listening to music, reading novels or going to the gym, the pleasure they give is what makes these tick. But if it is too much hard work, not many would pursue it. Those who enjoy electronics as a hobby are no different; putting together stuff and getting to see it in action is what draws them to it. In this article we introduce a tool that is tailored to suit the needs of electronics hobbyists.

Welcome to SimulIDE

SimulIDE is a real-time electronic circuit simulator. It is a simple tool intended for advance learning, and it lets you enjoy the experience as well. SimulIDE is designed to be fast and easy to use, and it works wonders for simple electronics. This is not your tool if your intention is to perform critical analysis or develop accurate models; it is ideal for trials and experimentation. Let us get down to the details of what this tool provides to you.

Readily-available controller models

Simple circuits are not just about putting together resistors, capacitors and inductors, of course. You often need a controlling section or an integrated circuit (IC). To aid easy development of such systems, SimulIDE provides support for simulation using microcontrollers.

SimulIDE provides AVR, Arduino and PIC microcontrollers that can be accessed just like other components. Features like gpsim and simavr allow you to use PIC and AVR

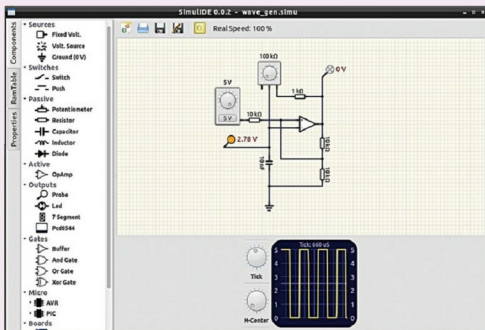


Fig. 1: A waveform generator implementation on SimulIDE (Image courtesy: <http://alternativeto.net/software/simulide/>)

Giving you the information you need

SimulIDE is an open source tool brought to you under GNU general-public licence. The entire project has been developed using C++, making use of Qt for the user interface. You can find the complete source code on their SourceForge page, <https://sourceforge.net/p/simulide>

If you have worked on Circuit Simulator or CircuitLab, you will find it very easy to get used to this tool, and get your simulation up and running quickly. If not, go ahead and check out the many video tutorials where simulations of small designs are demonstrated in great detail.

microcontrollers, respectively. Make sure these add-ons are installed on your system, too. You can load and reload firmware as many times as you want, and watch the movement of data between registers on the screen. The kit also comes with a serial terminal that you can pair with your board.

Design with ease

On getting into the application, a window opens up with a huge space

for the actual design. An all-powerful, right-click accessible context menu lets you select components and place these in the angle you want. You can even save the circuit as an image file at any instant easily from the same menu.

A column on the left-most corner shows three tabs. On opening any of these tabs; a column opens up between this and the design window. This acts as the user interface for the intricacies of